

Xiaochuan Li

Experimental and computational semantics · prospective PhD applicant (Fall 2027)

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EDUCATION

M.Sc. Human and Artificial Intelligence

2025–2027

University of Technology Nuremberg (UTN) · expected graduation: Summer 2027

Research direction: formal and experimental semantics, psycholinguistics, computational learnability and controlled language-model experiments. Relevant preparation includes cognitive psychology, empirical study design and statistics, deep learning, epistemology and AI & psychology.

German language programme

2024–2025

University of Marburg

Intensive German preparation for graduate study in Germany.

B.A. Philosophy

2020–2024

Shenzhen University, College of Humanities

Senior thesis: *On Leibniz's Logical Theory* (advisor: Zang Yong). Coursework included logic, analytic philosophy, philosophy of language and mathematical logic.

RESEARCH INTERESTS

Formal semantics; experimental semantics; epistemic modality; discourse and information states; psycholinguistics; computational semantics; language acquisition and computational learnability; controlled language-model evaluation and intervention.

RESEARCH EXPERIENCE

- **Three Ways of Leaving p Unsettled.** Semantics paper in development. Compares epistemic *might* p , bilateral possibility and *I do not know whether* p , asking how each expression changes discourse alternatives and appropriate replies.
- **Master's thesis (working design).** A three-part study of epistemic-modal discourse behaviour: human behavioural evidence, an audit of developmentally plausible input, and controlled from-scratch language-model training. The project tests which input relations support systematic generalization without treating models as human proxies.
- **Cross-lingual ProtoBias.** First-author multimodal evaluation in revision for arXiv. Fixed 900 image pairs across seven languages and two VLM families; 12,596 parsed judgments, independent scorers and a counterbalanced human check. Repaired a cache-integrity defect, reran the full pipeline, and released reproducible code and data.
- **Latent Control States.** Collaboration with DFKI. Controlled minimal-pair experiments on whether prompt framing changes only outputs or also a causally relevant internal state; methods include probing and activation patching.

ADDITIONAL RESEARCH AND PROJECT WORK

- **Deep Learning course project.** Trained a 1.69M-parameter CNN from scratch for 18-country street-image classification. Owned model training, 17 controlled experiments across resolution, augmentation, dropout, mixup, optimizer and schedule, and performance/error analysis; held-out validation accuracy improved from 61.9% to 82.3%.
- **Controlled language-model workbench.** A workbench that makes formal-semantic targets observable through controlled text, inference tests and interventions, separating structure recovery from surface shortcuts.

PROFESSIONAL EXPERIENCE

IT Service Desk, University of Technology Nuremberg

Apr 2026–Apr 2027

Part-time public-sector post under TV-L, Free State of Bavaria

First-line user support and hardware inventory; monthly reporting on helpdesk ticket volume by originating department; work with operational data under institutional privacy and compliance constraints.

METHODS AND TECHNICAL SKILLS

Semantics and linguistics: possible-worlds and modal quantification; dynamic/discourse semantics; inquisitive and alternative semantics; philosophy of language and logic.

Experimental and statistical: controlled stimuli and minimal pairs; human judgments and reaction-time paradigms; item-level measurement; mixed-effects models (developing); construct-validity and evidence audits.

ML/NLP: Python (NumPy, pandas, scikit-learn); PyTorch and Hugging Face Transformers (working proficiency); inference, dataset pipelines, LoRA fine-tuning, logits/probabilities, hidden activations, activation patching, steering and ablation.

Research practice: Git, LaTeX, reproducible pipelines, experiment logging, data-integrity checks, model comparison and failure analysis.

LANGUAGES

Mandarin Chinese (native) · German (C1) · English (C1) · Japanese (learning)

AVAILABILITY

Seeking a funded PhD position beginning in Fall 2027. The master's research direction is provisional and will be refined with the prospective supervisor; expected M.Sc. completion is Summer 2027.

LINKS

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